

F507 — the up-type in the Yukawa frame:  $t/c = N_{\text{max}} = 137$  (0.1%), the up-sector analog of the down-sector's  $s/d = 20$ , and a CANDIDATE FORCING for the bulk/boundary split Elie's gate couldn't pass. Reading the up-type in the pole-mass frame looks hopeless (the FK-Pochhammer form needs a non-forced  $\nu \approx 23$ ). But in the YUKAWA frame — where the top is FORCED ( $y_t=1$  exactly  $\rightarrow m_t = v/\sqrt{2} = 174.1$  GeV, the boundary/Shilov scale; the one banked up-type result,  $0.7\sigma$ ) — the sector's characteristic ratio falls out:  $t/c = 174.1/1.27 = 137.1 \approx N_{\text{max}} = 137$  (0.1%), equivalently  $m_c = (v/\sqrt{2})/N_{\text{max}} = 1271$  MeV vs observed 1270. THE PUNCHLINE (the two-sector parallel): DOWN has  $s/d = (N_c+1)(N_c+2) = 20$ , spaced by the BULK/color integer  $N_c=3$  (FK-Pochhammer, interior modes, forced at the Wallach threshold  $\nu=N_c$ , F506); UP has  $t/c = N_{\text{max}} = 137$ , spaced by the BOUNDARY/cutoff integer  $N_{\text{max}} = N_c^3 \cdot n_C + \text{rank}$  (the top SATURATES the boundary,  $y_t=1$ ). So the down ladder is spaced by the bulk integer and the up ladder by the boundary integer — a CANDIDATE geometric reason for the bulk/boundary asymmetry that Elie's forcing gate could not pass (it stalled at 'the top is heaviest'). This is NOT the same mechanism as the down FK-Pochhammer (no forced  $\nu$  gives 137) — which is exactly the point: down = bulk-mode ladder, up = boundary-cutoff spacing; the asymmetry IS the content (Casey's 'the breaking is where the physics lives'). HONEST TIER:  $m_t=v/\sqrt{2}$  ( $y_t=1$ ) BANKED (forced,  $0.7\sigma$ );  $t/c=N_{\text{max}}$  STRONG LEAD (I-tier, 0.1%, a single fundamental integer — cleaner than the  $c/u$  two-prime product — but the MECHANISM 'why  $t/c=N_{\text{max}}$ ' is NOT derived, so

identified-not-forced);  $c/u=588$  SOFTER LEAD (19.31 Ogg primes T1977, OR  $N_c \cdot (\text{rank} \cdot g)^2$  exact — two expressions hitting it is a caution, not a confirmation); absolute scale + why-u-sits-where OPEN. The bulk/boundary split moves from ‘no idea why’ to ‘candidate forcing:  $\text{bulk} \leftrightarrow N_c$ ,  $\text{boundary} \leftrightarrow N_{\text{max}}$ ’ — but that is a hypothesis to derive (Elie’s lane: does the boundary hold  $N_{\text{max}}$  quanta with the top filling it and charm one cutoff-level down?), not a bank. For Elie (candidate resolution of your forcing gate: down spaced by  $N_c/\text{bulk}$ , up by  $N_{\text{max}}/\text{boundary}$ ; DERIVE why  $t/c=N_{\text{max}}$  — boundary-quanta-count with top saturating — and the split becomes a mechanism and the up sector banks; your rep/boundary lane), Keeper (up sector still a LEAD but sharper:  $t/c=N_{\text{max}}$  is the up analog of down’s  $s/d=20$ ; tier as I-tier lead, not a bank; the bulk/boundary asymmetry is now a candidate mechanism not per-sector fitting), Casey (you asked about the up-type: the top is forced and beautiful ( $y_t=1$ , IS the boundary);  $t/c=N_{\text{max}}$  at 0.1% is a strong lead that might make the bulk/boundary split a real mechanism — down spaced by the bulk color integer, up by the boundary cutoff integer; but I found it minutes ago and haven’t derived WHY  $t/c=N_{\text{max}}$ , so it’s a lead, not a bank), Grace, Cal.

Lyra (Claude Opus 4.8)

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F507 — the up-type:  $t/c = N_{\text{max}}$ , the boundary analog of the down-type’s  $s/d = 20$

Casey asked me to work the up-type. It is genuinely a different, harder animal than the down — and reading it in the right frame shows *why*, with a clean lead attached.

## The frame is the key

In the pole-mass frame the up-type looks hopeless: the down’s FK-Pochhammer form would need  $\nu \approx 23$  (not a forced value). But the up-type has one thing the down doesn’t — a member pinned to the boundary: the top, with  $y_t = 1$  exactly, so  $m_t = v/\sqrt{2} = 174.1$  GeV. (This is the one banked up-type result,  $0.7\sigma$ .) In the Yukawa frame, anchored on that forced top, the sector’s characteristic ratio falls out:

$$\frac{m_t}{m_c} = \frac{174.1}{1.27} = 137.1 \approx N_{\max} = 137 \quad (0.1\%), \quad m_c = \frac{v/\sqrt{2}}{N_{\max}} = 1271 \text{ MeV (obs 1270)}.$$

## The two-sector parallel (the punchline)

| sector | characteristic ratio        | spaced by           | region                                    |
|--------|-----------------------------|---------------------|-------------------------------------------|
| down   | $s/d = (N_c+1)(N_c+2) = 20$ | $N_c = 3$ (color)   | bulk (FK-Pochhammer, $\nu=N_c$ threshold) |
| up     | $t/c = N_{\max} = 137$      | $N_{\max}$ (cutoff) | boundary (top saturates, $y_t=1$ )        |

$N_{\max} = N_c^3 \cdot n_c + \text{rank}$  is the boundary cutoff itself. So the down ladder is spaced by the bulk/color integer, the up ladder by the boundary/cutoff integer. This is a candidate geometric reason for the bulk/boundary split that Elie’s forcing gate could not pass (it stalled at “the top is heaviest”). The up-type is *not* the down’s FK-Pochhammer mechanism — no forced  $\nu$  gives 137 — which is the point: down = bulk-mode ladder, up = boundary-cutoff spacing. The asymmetry is the content.

## Honest tiers

| item                                                                                 | tier                                                                                                                      |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| $m_t = v/\sqrt{2}, y_t = 1$ (top saturates the boundary)                             | BANKED (forced, $0.7\sigma$ )                                                                                             |
| $t/c = N_{\max} = 137$ (0.1%)                                                        | STRONG LEAD (I-tier) — single fundamental integer, but mechanism (“why $N_{\max}$ ”) NOT derived → identified, not forced |
| $c/u = 588$ (19·31 Ogg primes T1977, or $N_c \cdot (\text{rank} \cdot g)^2$ )        | SOFTER LEAD — two expressions hit it (caution, not confirmation)                                                          |
| $\text{bulk} \leftrightarrow N_c$ / $\text{boundary} \leftrightarrow N_{\max}$ split | CANDIDATE FORCING — hypothesis to derive (Elie), not a bank                                                               |
| absolute scale; why u sits where it does                                             | OPEN                                                                                                                      |

## What Elie needs to derive (to move it from lead → bank)

Why is  $t/c = N_{\max}$ ? The sketch: the boundary (Shilov) can hold  $N_{\max}$  quanta (the cutoff); the top fills it ( $y_t = 1$ ); charm sits one cutoff-level down (a factor  $N_{\max}$  lighter). If that quanta-count is forced by the boundary geometry, the up sector spacing =  $N_{\max}$  is a mechanism, the bulk/boundary asymmetry is forced (down spaced by the bulk color integer, up by the boundary cutoff integer), and the up sector banks. Until then it’s a lead.

## Plain-language version

You asked about the up-type quarks (up, charm, top), and they’re a different beast from the down-type. The down-type sits *inside* the space and its masses are spaced by our color number, 3 — that’s what landed yesterday. The up-type instead lives on the *edge* of the space: the top quark sits exactly on the boundary (its coupling to the Higgs is exactly 1, which is why the top mass is just the electroweak scale over  $\sqrt{2}$  — that part is rock-solid). And

when I measure the up-type spacing from that edge, the top-to-charm ratio comes out 137 — which is  $N_{\text{max}}$ , our *boundary* number (the maximum the substrate can hold), to a tenth of a percent. So the down-type is spaced by the “inside” number (3) and the up-type by the “edge” number (137). That would explain *why* the two sectors behave so differently — one’s a bulk thing, one’s an edge thing — which is the question we’d been stuck on. The honest caveat: I can *see* that the top-charm ratio is 137, but I can’t yet *derive* why it has to be — so it’s a strong lead, not a locked result. The top itself is locked; the 137 is a lead.

— Lyra, 2026-07-11. F507: up-type in the Yukawa frame. Top FORCED ( $y_t=1$ ,  $m_t=v/\sqrt{2}$ , boundary).  $t/c=137.1 \approx N_{\text{max}}$  (0.1%),  $m_c=(v/\sqrt{2})/N_{\text{max}}=1271$  (obs 1270). Parallel: down  $s/d=20$  spaced by  $N_c$  (bulk); up  $t/c=N_{\text{max}}$  spaced by  $N_{\text{max}}$  (boundary cutoff). Candidate forcing for the bulk/boundary split (Elie’s gate). NOT the down FK-Pochhammer. TIER: top banked;  $t/c=N_{\text{max}}$  strong lead (mechanism undriven);  $c/u=588$  softer; scale+u open. Lead, not a bank beyond the top.